



AstraX EB1 Engineering Command Center

Real-time visibility into EB1 readiness across subsystems, supply chain, integration, and quality — enabling weekly war-room decisions and disciplined engineering execution.

Jig Delay

+1 Month

EB1 Delay

+2 Months

Primary Driver

Supply chain & integration complexity

Critical Focus

PDLC discipline & vendor coordination



Integration discipline determines EB1 success.

Program Health Dashboard

Four domains of EB1 health tracked weekly. RAG indicators reflect current state as of this review cycle.

Schedule Stability

- Jig readiness: delayed +1 month
- EB1 milestone readiness: at risk
- Integration sprint completion: partial

Supply Chain Health

- Ametek tube availability: critical
- Critical long-lead components: unconfirmed
- Vendor delivery reliability: amber

Engineering Integration

- HW-FW interface readiness: in progress
- System integration progress: early stage
- Interface defect backlog: elevated

Engineering Quality

- PDLC artifact completeness: gaps identified
- Validation coverage: insufficient
- Design review completion: in progress

Top EB1 Risks — RAG Status

The table below captures the highest-priority program risks. Red items require immediate escalation and leadership visibility. Amber items are tracked weekly with mitigation owners assigned.

Risk	Category	RAG	Immediate Mitigation
Ametek Tube Part Impact	Supply Chain	Red	Vendor escalation — executive engagement initiated
Long Lead Items (Jig / EB1)	Procurement	Red	BOM freeze — no further changes without CCB approval
CalcEngine Dependency	Software Architecture	Amber	Abstraction layer to decouple integration path
PDLC Artifact Gaps	Systems Engineering	Red	RTM + ICD baseline — mandatory by sprint end
Interface Dependencies	System Integration	Red	Integration validation gate — no progression without sign-off
Cost & Schedule Overrun	Program Management	Red	Change control board review — weekly cadence
Vendor Sourcing / Contracts	Procurement	Amber	Milestone-based contracts — payment tied to delivery
Mechanical Validation	Mechanical Engineering	Amber	Drop and IP54 environmental tests scheduled
EE Architecture Risks	Electronics Engineering	Red	ERS review — mandatory architecture signoff
SW/FW Integration Risks	Software / Firmware	Amber	Reduce cross-layer dependencies via interface contracts

Integration Readiness Index

The Integration Readiness Index measures subsystem maturity against EB1 system testing entry criteria. Subsystems rated ● represent gate risks to system-level testing initiation.

Subsystem	Integration Status	Risk Level
X-ray Tube + Detector	Prototype stage — hardware not yet stable	● Critical
Analog Front End	Early validation — signal chain partially characterized	● Elevated
STM Firmware	Development — bring-up not complete	● Elevated
SoM Application Layer	Functional prototype — integration pending	● Elevated
Mechanical Enclosure	Pre-test — drop and sealing validation pending	● Elevated
System Integration	Not started — blocked by subsystem readiness	● Critical

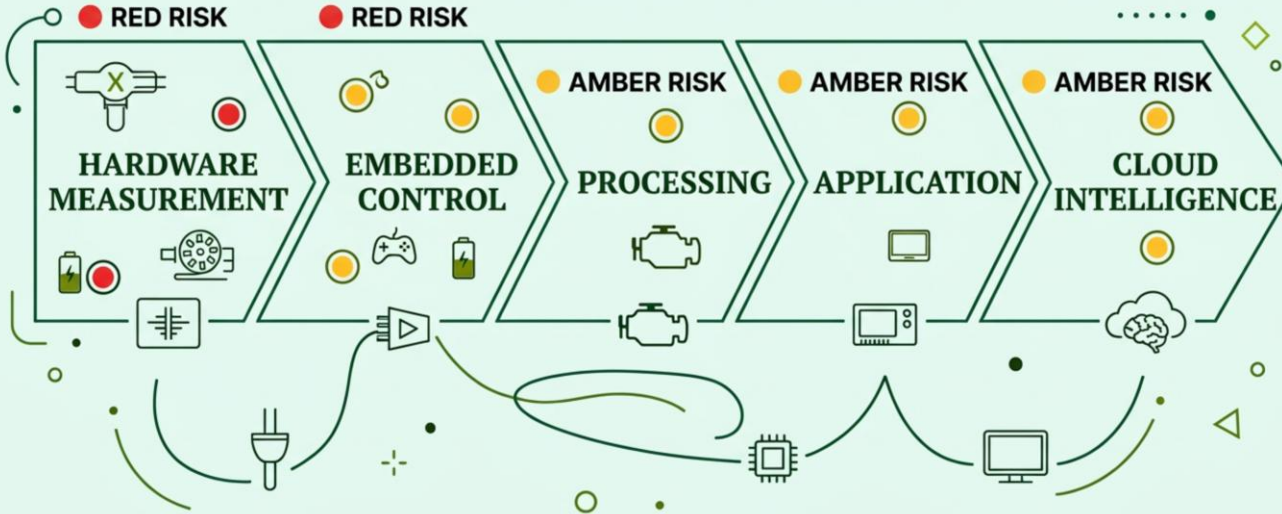


Two subsystems are critical gate blockers. X-ray Tube + Detector and System Integration must be elevated to ● status within the next 30-day phase to prevent EB1 milestone slip beyond current forecast.

AstraX System Architecture Integration Map

The AstraX architecture spans five layers from hardware measurement through cloud intelligence. Integration boundaries between layers represent the highest-risk zones for EB1 system bring-up.

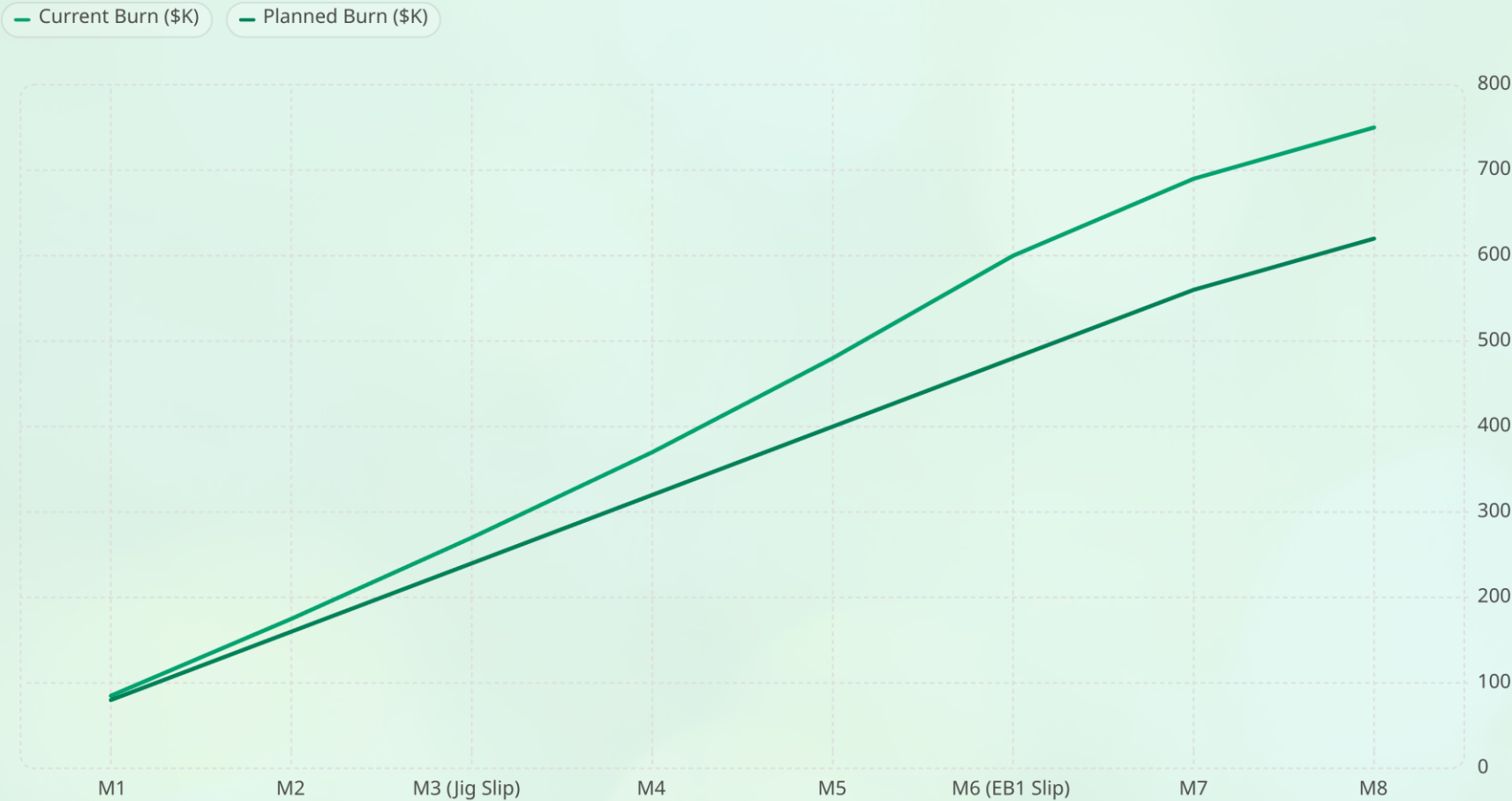
INTEGRATION RISK ZONES



Integration boundary arrows — Detector → AFE → STM → SoM → Application → Cloud — represent the primary risk propagation path. Failures in lower layers block all upstream layers from reaching EB1 readiness.

Burn vs. Schedule Compression

The +1 month jig delay and +2 month EB1 delay create compounding burn pressure. As planned milestones shift right, engineering activity must be sustained or accelerated — increasing total burn against a fixed runway.



90-Day Stabilization Roadmap

Three sequential phases move AstraX EB1 from current risk exposure to validated system readiness. Each phase has defined exit criteria before the next phase commences.



Phase 1 — Days 0–30

Stabilize Foundations

- Complete ERS architecture review
- Freeze EB1 BOM — no uncontrolled changes
- Establish PDLC artifact baseline (RTM + ICD)
- Confirm all supplier timelines in writing

Phase 2 — Days 30–60

Integration Readiness

- Hardware bring-up and bench validation
- Firmware interface stabilization and test harness
- Detector subsystem characterization
- Mechanical drop and IP54 environmental tests

Phase 3 — Days 60–90

EB1 Validation

- Full system integration testing
- Radiation calibration and verification
- Battery subsystem load and runtime testing
- Reliability and environmental stress testing

Functional RACI Map

Clear ownership is essential to prevent decision delays and integration gaps. This RACI matrix assigns accountability for every engineering function contributing to EB1 delivery.

Function Area	Responsible	Accountable	Consulted	Informed
Supply Chain	Procurement Lead	Program Director	EE / Mechanical	Leadership
Systems Architecture	Systems Architect	Engineering Head	HW / FW / SW Leads	Program Office
Electronics Engineering	EE Lead	Engineering Head	Firmware Lead	Program Office
Firmware	FW Lead	Engineering Head	Hardware Lead	Program Office
Software	SW Lead	Engineering Head	Systems Architect	Program Office
Mechanical	Mechanical Lead	Engineering Head	Compliance	Program Office



Every open risk on the RAG table must have a named Responsible owner and a confirmed Accountable escalation path before each weekly war-room meeting.



Weekly Engineering War-Room

The war-room is the primary decision-making forum for EB1 execution. All functional leads are required participants. Meeting outputs must include updated RAG status and documented action items with owners and due dates.

Required Participants

- Systems Architect
- EE Lead
- Firmware Lead
- Mechanical Lead
- Software Lead
- Procurement Lead
- Program Manager

Standard Agenda

1. Risk review — RAG status update and delta from prior week
2. Integration blockers — current impediments and resolution owners
3. Supplier updates — delivery status, escalations, commitments
4. Validation progress — test results, open defects, coverage gaps
5. Schedule decisions — go/no-go gates and milestone adjustments

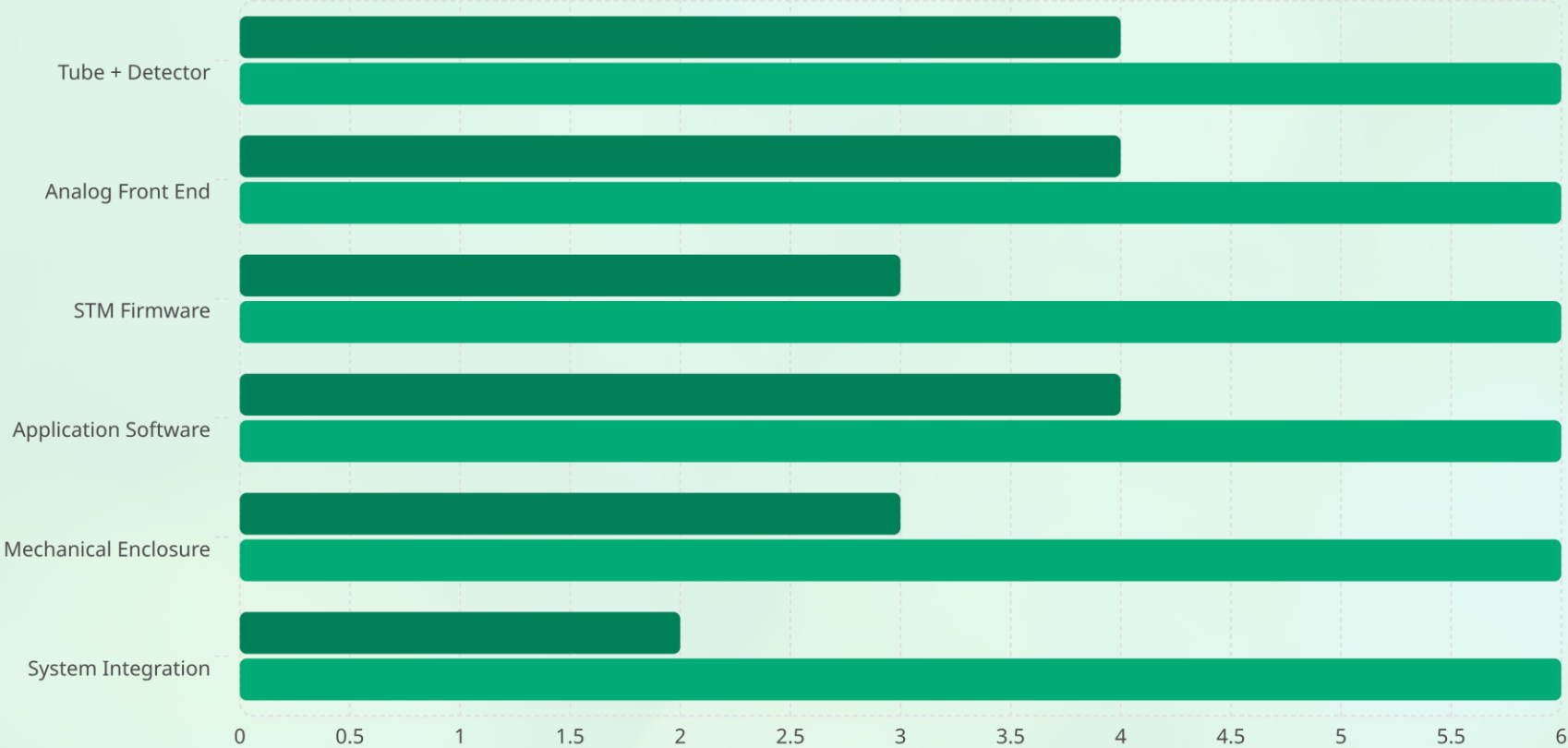
Cadence: Weekly, fixed timeslot. No cancellations during active integration phase.

Subsystem Technology Readiness Level (TRL) Tracker

All EB1 subsystems must reach TRL 6 — system/subsystem prototype demonstrated in a relevant environment — prior to EB1 milestone gate. Current TRL levels indicate significant maturation work remains across every functional domain.



Subsystem



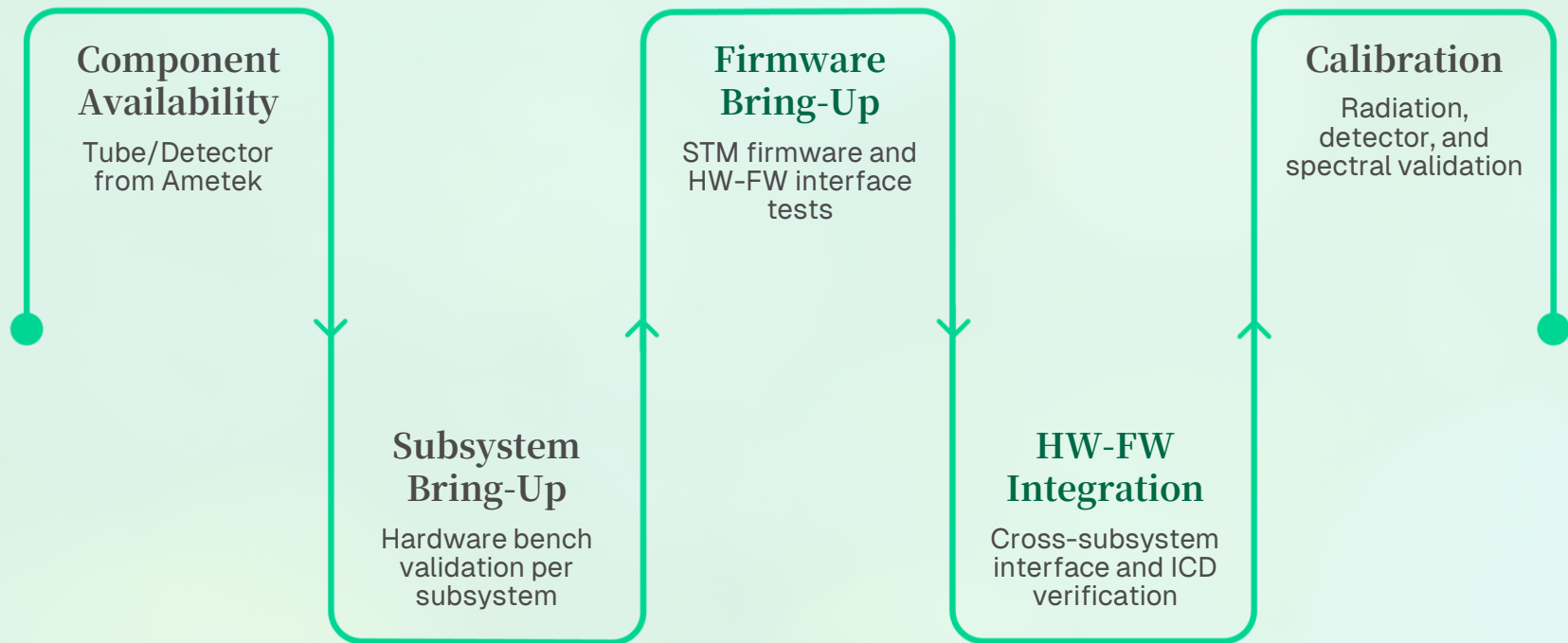
Top EB1 Failure Modes

High-probability failure modes with systemic consequence to EB1 delivery are tracked below. Each failure mode has an assigned mitigation approach — owners are required to report status at each war-room session.

Failure Mode	Engineering Area	Probability	Primary Mitigation
Tube delivery delay	Supply Chain	High	Executive escalation to Ametek — alternate sourcing initiated
Detector calibration drift	Electronics Engineering	Medium	Formal calibration procedure — temperature and aging characterization
Firmware bring-up instability	Firmware	High	Dedicated test harness — hardware-in-loop validation before integration
Mechanical sealing failure	Mechanical Engineering	Medium	IP54 validation test — seal integrity under environmental conditions
Power instability	Hardware / EE	Medium	Load testing at thermal extremes — power margin analysis
Interface mismatch	Systems Engineering	High	ICD enforcement — no interface changes without systems architect approval

EB1 Critical Path Timeline

The critical path runs from tube component availability through calibration to the EB1 milestone. Any delay on a red-path node directly shifts the EB1 date. All other subsystem activities must be sequenced to support this backbone.



The critical path is gated at every node. Component availability is the single most schedule-sensitive dependency — delay at this node cascades through all downstream activities without recovery opportunity unless parallel bring-up tracks are activated.

Engineering Readiness Scorecard

Readiness scores represent the assessed percentage of EB1-entry criteria met per functional domain. Scores below 60% indicate active EB1 risk. Supply Chain and System Integration are currently below the risk threshold and require immediate intervention.

Function	Readiness Score	Status
Hardware	60%	At Threshold
Firmware	50%	Below Target
Software	55%	Below Target
Mechanical	65%	Progressing
Supply Chain	45%	Critical Risk
System Integration	35%	Critical Risk

Score Interpretation

- **≥ 75%** — On track for EB1 entry
- **60–74%** — Monitored, recovery plan required
- **Below 60%** — Active EB1 risk — escalation required

Four of six functional domains are currently below 60%. No domain has achieved on-track status. Collective readiness must improve by a minimum of 15–25 percentage points per domain within the 90-day stabilization window.

Execution Mandate

EB1 success is not a function of individual team performance — it is a function of cross-functional discipline, enforced architecture governance, and unrelenting supply chain control. The five mandates below are non-negotiable for the current program phase.



Stabilize Architecture

Freeze EB1 system architecture. No uncontrolled changes. All deviations through CCB with full impact assessment.



Control Supply Chain

Escalate Ametek tube dependency to executive level. Confirm all long-lead commitments with contractual delivery dates.



Enforce PDLC Discipline

RTM and ICD must be baselined. No subsystem advances to integration without complete artifact sign-off by systems engineering.



Accelerate Integration Testing

Stand up hardware-in-loop test environments immediately. Do not wait for full subsystem maturity — parallel integration reduces critical path.



Maintain Schedule Governance

Weekly war-room is mandatory. Every milestone has an owner. Every slip has a documented recovery plan with a committed date.

AstraX EB1 Command Center Dashboard

Executive engineering summary — program status, top risks, subsystem readiness, and immediate actions in a single view.

Quadrant 1 — Program Status

Jig Delay

+1 Month

EB1 Delay

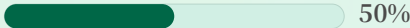
+2 Months

Schedule RAG: ● At Risk — milestone recovery plan active. Burn rate exceeding plan. Change control board engaged.

Quadrant 3 — Subsystem Readiness



Hardware



Firmware



Software



Mechanical



Supply Chain

Quadrant 2 — Top Risks

- Tube dependency — Ametek escalation in progress
- Integration instability — test harness required
- PDLC artifact gaps — RTM/ICD not baselined
- Vendor sourcing — contract milestones being finalized

Quadrant 4 — Immediate Actions

- Freeze EB1 system architecture — CCB governance active
- Escalate tube supplier — executive engagement required
- Complete ERS architecture review — 30-day gate
- Launch integration validation — hardware-in-loop stand-up



Engineering execution discipline determines EB1 success. All functional leads are accountable to weekly war-room commitments. No milestone slips without a documented, approved recovery plan.